Product recommendations and customer groups

To: Sarah Okafor, Head of E-Commerce **From:** Aditya Vishwanath, Data & ML Team **For:** Q4 planning meeting

In short: both things you asked for are ready. The new “also bought” suggestions find a product the customer really bought next **40% of the time, against 27% for a best-seller list**. The whole nightly job runs in **3.5 seconds** and stores about **4 MB**. For marketing, our customers fall into **six groups** defined by what they like to buy.

1. What we did and which method we chose

The old prototype compared every customer with every product: a table of 5,473 customers × 3,687 products, 161 MB when stored in full, even though 98% of it is empty because most customers buy only a few products. We used a method called **Truncated SVD**. It boils that table down to 50 “buying patterns”, such as “buys candle holders and lanterns” or “buys Regency tea sets”, and describes every customer and every product by those patterns. We also tested the alternative, **PCA**. It recommended equally well (40% vs 40%) but took three times as long and needs the full-size table again when making suggestions, so we kept SVD.

2. Is it better than showing our best sellers?

Yes. We hid the last two months of orders (9 Oct to 9 Dec 2011), built the model on the earlier data only, and checked whether each customer’s top 10 suggestions included a product they actually went on to buy for the first time. We tested this on 1,913 returning customers:

Approach	Customers with a correct suggestion in their top 10
Our model	**40%**
Top 10 best sellers (the simple alternative)	27%

The suggestions also make sense to a person: a white heart T-light holder suggests other T-light holders and lanterns, and a Regency cake stand suggests matching Regency teacups and a teapot. The model learned this from orders alone, without reading product names.

3. How big and how fast

Size: 161 MB for the full table, down to 3.7 MB for the 50 patterns plus 0.4 MB of ready-made top-10 lists. **Speed:** the whole nightly job (learning the patterns, building “also bought” lists for all 3,687 products and personal lists for all 5,473 customers) takes 3.5 seconds on a free cloud machine.

4. What we lose by compressing

The 50 patterns keep the main buying habits but drop individual one-off orders; in numbers, they keep about a third of the detail in the data. That loss does not hurt the suggestions, which beat the best sellers anyway. We also leave out products bought by fewer than 10 customers: they account for about 1 in 10 next purchases, and the model will not suggest them.

5. Customer groups for marketing

Vintage Tea Party (8%) · **Kids’ Crafts & Games** (19%) · **Cosy Comfort Gifts** (23%, smallest orders) · **Party Supplies** (14%, most customers outside the UK) · **Bags & Storage Bulk Buyers** (15%, largest orders) · **Romantic Home Décor** (22%). The groups overlap: many shops buy across several themes, so read each group as the customer’s *main* interest. Country separates them very little, because 81–98% of each group is UK-based. The map and summary table contain no customer IDs.

6. What could go wrong

New customers (16% of recent buyers) have no history, so they would see best sellers until they order. **Seasonality:** we tested on October–December, so results may be weaker in quieter months. **Repetitive suggestions:** lists often show variants of the same item (other cake cases), which may feel repetitive on a product page. **Settings chosen on the same test months:** the result may be slightly optimistic.

7. Recommended next steps

(1) Run a live A/B test of the box against best sellers on a share of product pages. (2) Limit lists to two items per product family. (3) Retrain monthly, and confirm the results on a separate set of months. (4) Show best sellers to new customers until they place a first order.